Lesson plan · for the teacher

IF/THEN: rules that act

IoT · Level 3 · Lesson 2

Time
40 min

Materials
Cards and markers for the rules; A day of readings on the board: time, temperature, light and tank level (or your log from lesson 1); Optional: a micro:bit, or a free simulator such as MakeCode or Wokwi, to run one rule with a real temperature sensor

Learning goals

- Write automation rules as IF condition THEN action, with sensible thresholds.
- Use two thresholds so devices don't flicker, and make the system fail safe.
- Test rules against a day of data and judge comfort against energy.

Activity: Rule cards through a school day

1. Warm-up (5 min): Ask: “When should our fan be on, and when is it wasting electricity?” Write pupils' ideas on the board as IF … THEN … sentences.
2. Teach (8 min): Show one rule with its threshold (fan on above 30 °C). Draw a wobbly line around 30 °C and ask what the fan does each time the line crosses it. Add the tank rule: pump on below 20 %.
3. Play the day (12 min): Three pupils hold the fan, light and pump cards. Read the day's readings hour by hour; each card-holder stands when their rule says on and sits when it says off. The class counts how often each device switches.
4. Fix it (10 min): Give the fan a second threshold (off only below 28 °C) and replay the wobbly stretch: count the switches again. Then read out “tank sensor: no reading” and agree what the pump should do: stop, and tell the caretaker.
5. Wrap-up (5 min): Each pupil writes one rule for a real device at school, with two thresholds and what happens if its sensor fails.

Before class, write a day of readings on the board or a chart (time, temperature, light, tank level), with a stretch that wobbles around 30 °C and one hour when the tank sensor is silent. Pupils act the devices: no mains appliances are switched in class.

Discussion

- Which school device would you automate first, and why?
- Who should choose the balance between comfort and electricity?
- What should happen when a sensor goes silent at night?

Exit ticket

- Write a two-threshold rule for the tank pump, and say what it does if the sensor goes silent.

Try it at school tomorrow
Rule cards through a school day: With your class, write three rules on cards (the fan, the lights and the tank pump), each with a threshold. Put a day of readings on the board (time, temperature, light and tank level), or use your log from lesson 1. Read it out hour by hour; pupils holding each card stand when their rule says on. Count how often each device switches, spot the flicker and fix it with two thresholds, then agree what each device should do if its sensor goes silent.

Practice questions · for you

IF/THEN: rules that act

IoT · Level 3 · Lesson 2

1 **The school's tank pump runs on "IF the tank is below 20 % THEN pump on". One morning the level sensor stops sending readings. What should the controller do?**

- ◯ Keep pumping, in case the tank is empty
- ◯ Stop the pump and alert the caretaker
- ◯ Run the pump for 10 minutes every hour until the sensor is back

2 **A rule that opens the windows when carbon dioxide passes 1,500 ppm keeps a classroom well ventilated.**

- ◯ True
- ◯ False

3 **A fan rule meets wobbly readings, one every 5 minutes: 29.8, 30.2, 29.9, 30.3, 30.1, 29.7 °C. The fan starts off. Count the switches.**

With one threshold (on above 30 °C, off at or below it), the fan switches _____ times. With two thresholds (on above 30 °C, off only below 28 °C), it switches on _____ time and stays on.

Word bank: 6 · 0 · 4 · 1

4 **In the Smart classroom rules lab, the fan rule starts at 29 °C. Which of these are true?**

- ☐ At 31 °C the fan never comes on, and lessons are too hot for over 2 hours
- ☐ Letting the lights burn when no class is in adds comfort
- ☐ Lowering it to 24 °C makes the class more comfortable
- ☐ At 30 °C there are still 0 minutes too hot, and less electricity is used

5 **Match each problem to its fix.**

The fan flicks on and off near 30 °C	• • • •	Raise the threshold, using logged data
The lights burn in the empty room at break	• • • •	Lower the CO_2 threshold towards 1,000 ppm
The pump keeps running when its sensor goes silent	• • • •	Add a condition: AND a class is in
The fan runs all morning on mild days	• • • •	A second threshold: off only below 28 °C
Stale air builds up before the window opens	• • • •	Fail safe: stop the pump and alert someone

✎ **Which device at your school could follow an IF/THEN rule? Write the rule with two thresholds, and say what it should do if its sensor fails.**

Check your answers

IF/THEN: rules that act

IoT · Level 3 · Lesson 2

1 **Stop the pump and alert the caretaker**

With no reading, the controller is blind.

2 **False**

Below about 800 ppm a room is well ventilated; at 1,500 ppm the air is already stale and pupils are getting sleepy, so that rule acts late.

3 **With one threshold (on above 30 °C, off at or below it), the fan switches 4 times. With two thresholds (on above 30 °C, off only below 28 °C), it switches on 1 time and stays on.**

With one threshold every crossing of 30 °C flips the fan: on at 30.2, off at 29.9, on at 30.3, off at 29.7, four switches in 25 minutes.

4 **At 30 °C there are still 0 minutes too hot, and less electricity is used; At 31 °C the fan never comes on, and lessons are too hot for over 2 hours**

From 29 or 30 °C the fan keeps the room from ever feeling over 30 °C, so 30 °C does the same job with less electricity (0.68 kWh a day against 0.75).

5 **The fan flicks on and off near 30 °C — A second threshold: off only below 28 °C**
The lights burn in the empty room at break — Add a condition: AND a class is in
The pump keeps running when its sensor goes silent — Fail safe: stop the pump and alert someone
The fan runs all morning on mild days — Raise the threshold, using logged data
Stale air builds up before the window opens — Lower the CO_2 threshold towards 1,000 ppm

Each fault has its own fix: flicker needs a gap between two thresholds, waste needs a better condition or threshold, and a silent sensor needs a safe default.